

Enhancing Human-Robot Teleoperation through Depth Sensing and eXtended Reality

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Figure 1: System architecture for teleoperation integrating the Reachy humanoid robot with a VR interface. The robot is controlled via a workstation running ROS and OpenCV, while Unity powers the VR environment with real-time AR overlays, such as object distances and bounding boxes. Bidirectional data flow allows the robot to send sensor data for visualization and the user to send control commands.

ABSTRACT

Teleoperation enables the control of robots in remote, complex, or hazardous environments such as manufacturing, healthcare, and disaster response. However, traditional teleoperation systems lack immersive depth perception, which limits spatial awareness and precision. This paper presents a novel teleoperation framework integrating an Intel RealSense D345i depth camera with the Reachy humanoid robot, enhanced by augmented reality (AR) overlays in a virtual reality (VR) interface. By leveraging depth sensing, point cloud processing, and AR visualization, the system improves task accuracy, spatial awareness, and user experience. Our system utilizes advanced frameworks for seamless AR and VR integration. Preliminary findings demonstrate enhanced performance in tasks requiring precise object manipulation, with future user studies planned for quantitative evaluation.

Index Terms: Human-centered computing—Human computer interaction (HCI)—User models—; Computing methodologies—Artificial intelligence—Robotics;

1 INTRODUCTION

Teleoperation enables users to remotely control robots, bridging physical distances and allowing for precise manipulation. These systems are crucial in scenarios where direct human involvement

is impractical or dangerous, such as hazardous material handling, remote healthcare, and disaster relief [1]. Despite advancements, traditional teleoperation systems often rely on 2D visual feedback, leading to a lack of depth perception that hinders performance in tasks requiring spatial precision [11].

This research addresses these challenges by enhancing telepresence through depth-sensing cameras and augmented reality (AR). Integrating an Intel RealSense D345i depth camera with a humanoid robot and leveraging AR overlays in a VR environment improves spatial awareness, task precision, and user interaction.

In this study, the Reachy humanoid robot, developed by Pollen Robotics, was utilized. Reachy is a highly modular and expressive robotic platform designed for advanced research in human-robot interaction [8]. Equipped with precise manipulators and compatible with various sensors, including depth cameras, it offers a robust foundation for exploring teleoperation systems in dynamic environments [13].

Leveraging humanoid robots in teleoperation scenarios addresses key challenges in remote operation, including limited feedback, precision, and adaptability [6, 7]. By combining these capabilities with AR/VR technologies, our system aims to improve both task performance and user experience.

The motivation and key contributions of this research include: The motivation for this research lies in overcoming the limitations of traditional teleoperation systems and advancing the field by integrating depth sensing and AR. Compared to prior studies that focus primarily on either VR or 2D interfaces, this work offers a novel approach by combining AR overlays with depth sensing in a VR environment for enhanced telepresence.

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The key contributions of this research include:

- Integration of depth sensing into a humanoid teleoperation system for real-time spatial feedback.
- Development of algorithms for depth data processing, object detection, and distance estimation.
- Implementation of AR overlays in a VR interface to improve telepresence and spatial awareness.

This research not only enhances task performance but also explores the potential for real-world applications, including industrial automation, remote healthcare, and disaster management. By addressing the limitations of traditional systems, the proposed framework sets the foundation for more immersive, efficient, and adaptable teleoperation systems.

2 RELATED WORK

2.1 Traditional Teleoperation Challenges

Teleoperation systems have been widely used in domains such as manufacturing, healthcare, and disaster management, where remote operation of robots is essential. These systems traditionally rely on standard 2D video feeds transmitted from robot-mounted cameras to the operator's console. While effective for basic monitoring, these setups fail to provide adequate depth perception, which is crucial for tasks involving precise spatial manipulation, such as picking, assembling, or navigating cluttered environments.

The lack of depth perception requires operators to rely heavily on visual cues and prior experience to estimate distances, increasing **cognitive load** and making tasks more **time-consuming** and **error-prone** [16]. For instance, objects may overlap or move unpredictably in cluttered or dynamic environments, further complicating task execution. Studies have shown that such limitations lead to:

- Increased task completion times.
- Higher rates of operational errors.
- More significant operator fatigue is due to prolonged focus and cognitive strain [11].

Despite improvements in camera quality and processing power, these challenges remain a fundamental limitation of 2D video-based teleoperation systems.

2.2 Depth Sensing for Robotics

The advent of depth-sensing technologies has revolutionized spatial data acquisition in robotics [5]. Devices like the Intel RealSense depth cameras utilize stereo vision, active infrared (IR) sensing, and structured light to capture detailed 3D information about the environment. This depth data enhances traditional 2D imagery by providing spatial context, enabling robots to understand and interact with their surroundings more effectively.

Key applications of depth-sensing in robotics include:

- **Object Detection and Segmentation:** Depth data accurately identifies objects, even in complex or overlapping scenarios. Algorithms such as RANSAC (Random Sample Consensus) and Euclidean clustering segment objects from the background using Point Cloud Library (PCL) [3, 14].
- **Navigation and Path Planning:** Depth maps enable robots to navigate safely by detecting obstacles and planning collision-free paths in real-time.

- **Grasping and Manipulation:** By providing precise distance measurements, depth sensors improve the accuracy of robotic arms in picking, holding, and placing objects.

Integrating depth-sensing into teleoperation systems has shown promising results in reducing operator workload and enhancing task accuracy. However, challenges remain in synchronizing depth data with real-time feedback systems, as delays can negatively affect operator performance.

2.3 Augmented Reality in Teleoperation

AR technologies have emerged as a powerful tool to enhance spatial awareness in teleoperation systems. By overlaying real-time information such as object locations, distances, and robot status onto a user's field of view, AR systems provide an intuitive and immersive interface for controlling robots [10].

Key benefits of AR in teleoperation include:

- **Improved Spatial Awareness:** Operators gain a better understanding of the robot's environment through visual overlays, such as bounding boxes, proximity indicators, and path predictions [2, 15].
- **Reduced Cognitive Load:** AR simplifies complex tasks by presenting critical information directly within the operator's line of sight, eliminating the need to switch between different data sources or displays [7].
- **Enhanced Collaboration:** AR interfaces enable multiple operators to share and interact with the same augmented environment, improving coordination in multi-robot or team-based scenarios [17].

Recent advancements in AR hardware, such as the Meta Quest Pro VR headset, have made integrating AR with teleoperation systems feasible. These devices support multimodal interaction through features such as voice recognition, hand tracking, and eye tracking, enabling operators to interact with robots in intuitive and efficient ways [12]. High-resolution displays and accurate head tracking ensure that AR overlays remain stable and responsive during operation. When combined with depth sensing, AR further enhances the operator's ability to perceive and interact with the remote environment by providing real-time depth visualization, object labeling, and trajectory planning.

For example, Omarali et al. [10] demonstrated a virtual reality-based telerobotics framework that integrates depth cameras with AR to enhance object manipulation tasks. Their system utilized color-coded proximity indicators, such as green for safe zones and red for danger zones, enabling operators to make quick and informed decisions during critical operations. Similar techniques could be extended to collaborative multi-robot environments for improved coordination and safety. Color-coded proximity indicators (e.g., green for safe, red for close) in AR interfaces help operators make quick, informed decisions during critical operations.

Despite these benefits, the integration of AR into teleoperation systems introduces challenges such as:

Latency: Maintaining low-latency updates for AR overlays is critical for real-time responsiveness. Techniques such as predictive rendering or edge computing can help mitigate this issue.

Complexity: Synchronizing AR data with depth sensing and robot control systems in dynamic environments requires robust architectures and real-time synchronization protocols to ensure seamless operation.

2.4 Comparison with Similar Systems

Existing systems leveraging AR and depth sensing for teleoperation, such as those by [10, 18], have primarily focused on improving spatial awareness and reducing cognitive load. However, these systems often lack comprehensive integration with humanoid robots or fail to address the real-time synchronization challenges in dynamic environments. Compared to [18], who integrated depth maps into 2D interfaces, our system advances this approach by providing immersive AR overlays in a VR environment. Our research uniquely combines depth sensing, AR overlays, and a humanoid robot platform (Reachy) to deliver real-time depth visualization, intuitive feedback, and enhanced user interaction. This integrated approach addresses key gaps in existing literature, offering a robust framework for improving task precision and operator efficiency in teleoperation.

3 SYSTEM DESIGN AND IMPLEMENTATION

3.1 Hardware Integration

The system's hardware architecture is designed to capture real-time depth data and synchronize it with the humanoid robot's movements, ensuring a seamless teleoperation experience.

Intel RealSense D345i Depth Camera:

- **Depth Data Capture:** The RealSense D345i uses a combination of stereo vision and active infrared (IR) technology to generate precise depth maps, even in low-light conditions. It captures depth data with a resolution of up to 1280×720 pixels at **30 frames per second (fps)**, providing real-time spatial awareness.
- **Camera Mounting:** The camera was mounted on the robot's head to mimic human visual perception, allowing it to cover a wide field of view and reduce blind spots during teleoperation tasks (Figure 2).

Synchronization with Reachy's Motor Control: ROS (Robot Operating System) nodes were developed to integrate the camera's depth data into the robot's control pipeline. This required:

- Establishing a **sensor fusion mechanism** to combine depth data with Reachy's joint position data.
- Implementing real-time **data buffering and timestamp alignment** to minimize latency, reducing it to **200 ms**.

Data Transmission: Depth data was transmitted to the AR interface via a wireless network. The system used Wi-Fi 6 to ensure high-bandwidth, low-latency communication between the robot and the AR headset.

3.2 Depth Data Processing

Processing the raw depth data is crucial for extracting actionable spatial information for teleoperation tasks. The processing pipeline includes the following steps:

- **Noise Filtering:** Depth data often contain artifacts due to environmental interference (e.g. reflective surfaces or occlusions). A voxel grid filter from the Point Cloud Library (PCL) was used to downsample the point cloud and reduce noise without compromising accuracy.
- **Object Segmentation:**
 - **Euclidean Clustering** This algorithm was implemented to segment objects based on spatial proximity within the point cloud. It grouped points into clusters, each representing a distinct object [4].



Figure 2: Reachy humanoid robot equipped with an additional camera for enhanced teleoperation.

- **Cluster Thresholds** Tunable parameters were set to define minimum and maximum cluster sizes, improving segmentation in cluttered environments.
- **Distance Estimation :** The centroid of each segmented object cluster was calculated to determine its distance from the robot. This enabled for precise object localization relative to Reachy.
- **3D Bounding Box Generation:** The Minimum Volume Bounding Box algorithm was applied to ensure a consistent spatial representation. This method calculates the smallest 3D box that encloses an object, which is then visualized in the AR interface.

The depth data processing pipeline achieves a frame processing time of 0.8 s, ensuring compatibility with real-time teleoperation requirements.

3.3 AR Interface Design

The AR interface was designed to provide intuitive and immersive feedback to the operator using a Meta Quest Pro VR headset. Key features include:

- **Depth Visualization:** Real-time depth data is rendered as visual overlays on the AR display. The detected objects are enclosed in bounding boxes with labels showing their distance from the robot.
- **Proximity Indicators:** Objects are color-coded based on their distance from Reachy:
 - **Green:** Objects that are safe from the robot arms.
 - **Yellow:** Objects within an intermediate range.
 - **Red:** Objects in close proximity, signaling potential collision risks.

These indicators dynamically adjust as the robot or objects move.

- **User Interface Enhancements:**

- **Distance Awareness Mode:** This mode highlights the closest objects within a user-defined range, helping operators prioritize tasks in complex environments [9].
- **User Calibration:** The AR system includes a calibration step to align the headset's perspective with the camera's field of view, ensuring accurate overlays.

- **Implementation Details**

- **OpenXR Framework:** Provides cross-platform support for AR development, enabling compatibility with various VR and AR headsets.
- **Unity3D Integration:** Unity was used to handle rendering, user interaction, and low-latency updates. The system achieved an average rendering latency of **200 ms**, ensuring responsive feedback.

The integration of hardware, depth sensing, and the AR interface is depicted in Figure 1, which illustrates the system architecture for teleoperation.

4 RESULTS AND FINDINGS

4.1 Object Detection Performance

A custom dataset of 5,000 labeled images of various shapes, such as cubes, spheres, and cylinders, was used to train the object detection model. The model achieved:

- **Detection Accuracy:** **78%** accuracy in identifying objects across different lighting conditions and partial occlusions.
- **False Positive Rate:** **12%**, primarily in overlapping objects or poor lighting.
- **Inference Time:** The model processes a single image in **0.8 s** on average, suitable for real-time operation.

4.2 Observations and Challenges

Preliminary tests were conducted by the authors to validate the functionality and performance of the integrated teleoperation system. These tests focused on object manipulation tasks, with key findings as follows

- **Improved Task Performance:** The time required for grasping objects decreased using the integrated system. The success rate of precise task completion also increased.
- **Data Synchronization:** Initial latency in synchronizing depth camera data with Reachy sensors was measured at 2 s. By optimizing ROS nodes and refining the sensor pipeline, this latency was reduced to 500 ms.
- **Real-Time Feedback:** Initial latency in synchronizing the depth camera data with Reachy sensors was measured at 2 s. By optimizing the ROS nodes and refining the sensor pipeline, this latency was reduced to 500 ms.

Methodology: The Reachy robot was tasked with grasping and relocating objects of different shapes and sizes. Metrics like task completion time and task accuracy were recorded.

4.3 Future Expectations

It is anticipated that the integration of a custom-trained AI model with depth sensing and AR will result in a significant enhancement in decision-making accuracy, achieving an improvement of 15–20%. This synergy between advanced AI and immersive technology is expected to empower users with more precise and reliable insights, facilitating better decision-making processes.

Additionally, the system's ability to provide real-time feedback is projected to lead to further optimization of task completion times, improving them by 10–15%. This improvement will directly contribute to increased operational efficiency, making the system more effective for practical, real-world applications.

5 USER STUDY DESIGN

To validate the proposed teleoperation system, a user study will be conducted to evaluate its performance, usability, and overall effectiveness. A diverse group of participants, including both novice and experienced users in teleoperation, will be recruited to ensure comprehensive feedback. The study will involve at least 20 participants to achieve statistically significant results, with a balanced representation of experience levels.

Participants will engage in tasks designed to mimic real-world teleoperation scenarios. These tasks will include object grasping, where participants will grasp objects of varying shapes and sizes—such as spheres, cubes, and cylinders—placed at different distances. Another task, object positioning, will require participants to place the grasped objects precisely in specified locations, such as aligning them within a target zone. Each task will be performed under two conditions: using a traditional 2D feedback system (without AR integration) and the enhanced AR-integrated setup.

The purpose of the study is to assess the impact of AR-enhanced depth visualization on task performance and user experience compared to traditional 2D interfaces. It will also aim to identify usability challenges and gather participant feedback to refine the system further.

Several metrics will be used to evaluate the system's effectiveness. Task completion time will measure the duration required for participants to complete each task under both conditions. Task accuracy will be assessed as the percentage of successfully completed tasks, such as accurately grasping and positioning objects or avoiding obstacles. Additionally, user satisfaction will be measured using a Likert-scale survey (ranging from 1 to 5), capturing participants' perceptions of ease of use, intuitiveness, and overall satisfaction with the system.

This user study will provide both quantitative and qualitative insights into the effectiveness of the AR-enhanced teleoperation system. The findings will help refine its design and functionality, paving the way for its application in real-world scenarios.

6 CONCLUSION AND FUTURE WORK

This work demonstrates the potential of integrating depth sensing and AR to enhance telepresence in human-robot interaction. By combining the Intel RealSense D345i depth camera with the Reachy humanoid robot and leveraging AR overlays in VR, we have addressed critical limitations of traditional teleoperation systems, such as the lack of depth perception and spatial awareness. This system provides real-time depth visualization, object detection, and intuitive feedback, enabling users to perform tasks with greater precision and efficiency.

Future work aims to build upon this foundation in several key areas. First, advanced machine learning techniques will be integrated to further refine the system's capabilities. Object detection algorithms will be trained on larger and more diverse datasets to improve accuracy and reduce false positives. Additionally, reinforcement learning methods will be incorporated to optimize robot control strategies and adapt to dynamic environments. The system's

applicability to real-world scenarios will also be explored. It will be adapted for use in settings such as industrial assembly lines, remote surgery, and disaster management. Rigorous testing in dynamic environments will be conducted to ensure robustness in complex tasks, including obstacle navigation and multi-object manipulation. Finally, the collaborative dynamics between human operators and robots will be studied in detail to identify synergies that enhance team efficiency. Understanding these interactions will be crucial for fostering effective human-robot collaboration in future applications.

REFERENCES

- [1] D. Aschenbrenner, M. Fritscher, F. Sittner, M. Krauß, and K. Schilling. Teleoperation of an industrial robot in an active production line. *IFAC-PapersOnLine*, 48(10):159–164, 2015.
- [2] M. Billinghurst and H. Kato. Collaborative augmented reality. *Communications of the ACM*, 45(7):64–70, 2002.
- [3] P. F. Felzenszwalb and D. P. Huttenlocher. Efficient graph-based image segmentation. *International journal of computer vision*, 59:167–181, 2004.
- [4] R. Girshick. Fast r-cnn. *arXiv preprint arXiv:1504.08083*, 2015.
- [5] K. Khoshelham and S. O. Elberink. Accuracy and resolution of kinect depth data for indoor mapping applications. *sensors*, 12(2):1437–1454, 2012.
- [6] J. Lee, S. Miri, A. Bayro, M. Kim, H. Jeong, and W.-H. Yeo. Biosignal-integrated robotic systems with emerging trends in visual interfaces: A systematic review. *Biophysics Reviews*, 5(1), 2024.
- [7] B. Marques, S. Silva, E. Pedrosa, F. Barros, A. Teixeira, and B. S. Santos. Towards unlimited task coverage and direct (far-off) manipulation in extended reality remote collaboration. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, pp. 745–749, 2024.
- [8] S. Mick, M. Lapeyre, P. Rouanet, C. Halgand, J. Benois-Pineau, F. Paclet, D. Cattaert, P.-Y. Oudeyer, and A. de Rugy. Reachy, a 3d-printed human-like robotic arm as a testbed for human-robot control strategies. *Frontiers in neurorobotics*, 13:65, 2019.
- [9] P. Milgram and F. Kishino. A taxonomy of mixed reality visual displays. *IEICE TRANSACTIONS on Information and Systems*, 77(12):1321–1329, 1994.
- [10] B. Omarali, B. Denoun, K. Althoefer, L. Jamone, M. Valle, and I. Farkhatdinov. Virtual reality based telerobotics framework with depth cameras. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, pp. 1217–1222. IEEE, 2020.
- [11] G. Pang, G. Yang, and Z. Pang. Review of robot skin: A potential enabler for safe collaboration, immersive teleoperation, and affective interaction of future collaborative robots. *IEEE Transactions on Medical Robotics and Bionics*, 3(3):681–700, 2021.
- [12] K.-B. Park, S. H. Choi, J. Y. Lee, Y. Ghasemi, M. Mohammed, and H. Jeong. Hands-free human–robot interaction using multimodal gestures and deep learning in wearable mixed reality. *IEEE Access*, 9:55448–55464, 2021.
- [13] P. Robotics. This is reachy. Retrieved March 24, 2024, from <https://www.pollen-robotics.com/>.
- [14] R. B. Rusu and S. Cousins. 3d is here: Point cloud library (pcl). In *2011 IEEE international conference on robotics and automation*, pp. 1–4. IEEE, 2011.
- [15] M. Sereno, X. Wang, L. Besançon, M. J. McGuffin, and T. Isenberg. Collaborative work in augmented reality: A survey. *IEEE Transactions on Visualization and Computer Graphics*, 28(6):2530–2549, 2020.
- [16] T. Sheridan. *Telerobotics, Automation, and Human Supervisory Control*. MIT Press, 1992.
- [17] P. Wang, X. Bai, M. Billinghurst, S. Zhang, X. Zhang, S. Wang, W. He, Y. Yan, and H. Ji. Ar/mr remote collaboration on physical tasks: a review. *Robotics and Computer-Integrated Manufacturing*, 72:102071, 2021.
- [18] Z. Zhang. Iterative point matching for registration of free-form curves and surfaces. *International journal of computer vision*, 13(2):119–152, 1994.